

SRAI

BOOK 2 · CHAPTER 1

Supervised Learning Foundations

Audited Controlled Review Candidate · Version 0.3

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Publication Control Note

CONTROLLED STATUS Complete SRAI rewrite aligned to the structure and teaching method of Book 1 Chapter 1. Owner review remains required before downstream asset production or public release.

Book: Book 2 — Machine Learning

Chapter: Chapter 1 — Supervised Learning Foundations

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Canonical notebook: M2_N01_supervised_learning_foundations.ipynb

Notebook validation: PASS; seed 42; regression test RMSE 30.876; classification test ROC AUC 0.955

Related formats: GitHub package, website lesson page, LinkedIn educational post and YouTube lesson

Scope of this controlled rewrite

- Reconstruct the chapter in the canonical SRAI sequence used in Book 1 Chapter 1.
- Preserve mathematical rigor while keeping the decision problem visible.
- Align every worked result with the executed companion notebook.
- Separate verification, validation, interpretation and decision.
- Replace generic expansion material with chapter-specific reasoning.
- Provide progressive exercises, a reproducibility checklist, assessment and solutions.

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2.1 From Labelled Data to Reliable Predictions

Controlled alignment: Book 2 master source and executed companion notebook

Canonical companion: M2_N01_supervised_learning_foundations.ipynb

A predictive model becomes scientifically useful only when we can explain what it learns, reproduce how it was evaluated, and show that its predictions remain credible on cases it did not see.

2.2 Learning Objectives

By the end of this chapter, the reader should be able to:

1. Define observations, features, targets, labels, models and loss functions.
2. Translate a real decision need into a supervised learning task.
3. Explain generalization and distinguish it from training fit.
4. Assign distinct roles to training, validation and test data.
5. Recognize underfitting, overfitting and common forms of leakage.
6. Compare a learned model with credible baselines and appropriate metrics.
7. Build a reproducible experiment in which preprocessing remains inside the training process.
8. Evaluate an AI-generated model claim rather than accepting it because the score appears impressive.

2.3 Opening Story

2.3.1 The Model That Knew Tomorrow

A national service agency wants to anticipate next month's demand at each facility. The objective is practical: move temporary staff before long queues appear.

The analytical team assembles facility records, staffing levels, calendar indicators and past visits. A model reports extraordinary performance. Its error is almost zero. The result is circulated rapidly.

Managers begin planning deployment.

Then a second analyst reconstructs the data pipeline.

One feature, *resolved_count*, is finalized only after the month has ended. It is almost the same quantity as the target. The preprocessing step was fitted before the data were divided. Future months helped determine transformations applied to earlier months. Records from the same facilities appeared throughout the development and evaluation samples.

The model did not discover tomorrow.

Tomorrow had entered the data.

The score still exists.

The evidence supporting it does not.

This story illustrates a central principle of SRAI:

A prediction is not yet evidence of generalization.

Evidence requires a defensible chain connecting the decision problem, prediction time, data-generating process, information boundary, model, evaluation design, computation, checks, interpretation and action.

2.4 The Central Question

How do we learn from labelled examples without confusing performance on known data with reliable prediction on unseen cases?

This question is more important than any particular algorithm.

Linear regression, decision trees, support-vector machines, neural networks and foundation models can all fit data. The scientific problem is to determine what the fitted relationship means, whether the evaluation reproduces the intended use, and what evidence supports action.

A fitted model is therefore not knowledge.

It is a candidate rule whose credibility must be tested.

2.5 Observations Features Targets and Prediction Time

Suppose an agency predicts facility demand one month ahead.

One observation may represent one facility-month. The feature vector may contain:

- visits during earlier months;
- catchment population;
- staffing known at the forecast date;
- season and calendar indicators;
- planned closures already approved;
- geographic or facility characteristics.

The target is next month's finalized number of visits.

For observation i , supervised learning represents the labelled example as

$$\text{left}(x_i, y_i \text{right}),$$

where $x_i \in X$ is the feature vector and $y_i \in Y$ is the target.

The vector is not the facility.

It is a selected representation of information available at a declared prediction time.

Every representation includes both knowledge and omission. Supervised learning begins when we ask:

- What does one row represent?
- When is the prediction issued?
- Which outcome occurs after that time?
- Which features genuinely exist at that time?

- Which populations and periods does the sample represent?
- What action will follow the prediction?
- What evidence could show that the system is inadequate?

2.6 The Anatomy of a Supervised Learning System

Most useful supervised learning systems contain six elements.

2.6.1 Population and observations

The population identifies the cases to which the model will be applied. Observations are the recorded examples available for learning.

Operational databases are not automatically representative samples. Records may reflect access, reporting, institutional rules, missingness, previous interventions and changing definitions. Repeated observations from the same household, facility, enterprise, district or person create dependence that the split design must respect.

2.6.2 Features and targets

Features are inputs available at prediction time. The target is the outcome the model is intended to predict. A label is the recorded operational version of that target.

Confusing target and label can hide measurement error. Confusing current information with future information creates leakage. A database column is not a valid feature merely because it exists when the analyst downloads the data.

2.6.3 Hypothesis class and fitted model

A learning algorithm selects a function

$$\hat{f}: X \rightarrow Y$$

from a set of candidate functions. This set is the hypothesis class. Its capacity determines the range of patterns the model can represent.

A fitted model is conditional on its data, features, objective, algorithm, hyperparameters and random choices. It should never be described as though it appeared independently of these decisions.

2.6.4 Loss and objective

A loss function measures the cost of a prediction error. Empirical risk is

$$\hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)).$$

Training often selects parameters that minimize empirical loss, possibly with regularization:

$$\hat{f} = \underset{f \in F}{\operatorname{argmin}} [\hat{R}_n(f) + \lambda \Omega(f)].$$

The mathematical objective reflects values. Squared error emphasizes large misses. Classification accuracy may ignore a rare but consequential class. The optimization objective must remain connected to the real decision objective.

2.6.5 Generalization

The quantity of interest is expected loss on relevant future cases:

$$R(f) = E[L(Y, f(X))].$$

The difference between future risk and observed training risk is the generalization gap. Low training error does not guarantee low future error.

Underfitting occurs when the model cannot capture relevant structure. Overfitting occurs when it learns noise, accidental detail or development-specific choices. Model capacity, sample size, measurement quality, representativeness and regularization all influence generalization.

2.6.6 Evidence

A model must confront evidence that reflects its intended use.

Evidence may include untouched test performance, baseline comparisons, learning curves, residual diagnostics, calibration, subgroup assessment, uncertainty, sensitivity analysis, temporal or geographic transfer, and expert review.

Computation fits a model.

Evidence helps determine whether the model is useful.

2.6.1 Professional Deepening 1 Purpose and learning contract

Supervised learning turns labelled observations into predictions for cases the model has not seen. The central discipline is therefore not curve fitting but controlled generalization. This lesson develops the conceptual, mathematical, computational, and governance foundations needed to judge whether a predictive system is credible.

By the end, the learner should be able to translate an operational question into a prediction problem; distinguish features, targets, observations, and prediction times; explain empirical and expected risk; construct training, validation, and test partitions; detect common forms of leakage; use baselines and learning curves; select metrics that reflect decision costs; quantify uncertainty; and define the evidence required before deployment.

The lesson uses a public-service demand case because it makes the temporal and institutional dimensions explicit. The methods apply equally to agriculture, finance, health, education, logistics, and enterprise systems. Throughout, model quality is treated as one component of decision quality rather than as an isolated numerical score.

2.6.1.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
- Which information is unavailable, selectively observed, delayed or generated after the outcome?
- Which alternative design would give a more credible estimate of generalization?
- What baseline, uncertainty measure and subgroup evidence must accompany the headline metric?
- What result would cause the analyst to stop, redesign the experiment or restrict intended use?

2.6.2 Professional Deepening 2 From decision need to prediction task

A useful model begins with a decision, not with an algorithm. The sponsor may ask which facilities will face unusually high demand, which farms are at risk of crop loss, or which transactions warrant review. Each request must be converted into a unit of analysis, prediction horizon, target, available information set, action, and error-cost statement.

The unit of analysis identifies what one row represents. The prediction time states when the forecast is issued. The horizon states how far ahead the target occurs. These definitions prevent future information from entering the features. They also expose whether labels are delayed, selectively observed, or altered by previous interventions.

A prediction is valuable only when someone can act on it. If an agency cannot change staffing within thirty days, a one-week demand forecast may be accurate yet operationally irrelevant. The evaluation design must therefore connect the model output to a feasible action and to the consequences of false positives, false negatives, large numeric errors, and abstention.

2.6.2.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
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2.6.3 Professional Deepening 3 Formal supervised learning framework

Let each labelled observation be a pair

$$(x_i, y_i)$$

drawn from an operating population. The feature vector

$$x_i \in X$$

contains information available at prediction time, and the target

$$y_i \in Y$$

records the outcome to predict. A learning algorithm uses a training sample to select a function

$$\hat{f}: X \rightarrow Y$$

from a hypothesis class.

A loss function

$$L(y, \hat{f}(x))$$

assigns a cost to a prediction. Training commonly minimizes empirical risk,

$$\hat{R}_n(f) = \frac{1}{n} \sum_{i=1}^n L(y_i, f(x_i)),$$

possibly with a regularization term. The quantity of real interest is expected risk on future cases,

$$R(f) = E[L(Y, f(X))].$$

The gap between empirical and expected performance is the generalization gap.

The mathematical notation is compact, but every symbol encodes a design choice. The population determines the meaning of the expectation; the feature space determines what is knowable at prediction time; the target determines what success means; and the loss determines which errors matter. A technically correct optimizer cannot repair a poorly specified target or an unrepresentative sample.

2.6.3.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
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2.6.4 Professional Deepening 4 Regression classification and other target structures

Regression predicts a continuous target such as monthly demand, yield, price, or processing time. Classification predicts categories, often with estimated probabilities. Binary classification might estimate the probability that demand exceeds capacity; multiclass classification might assign a case to one of several service pathways.

The same substantive question can sometimes be framed in more than one way. Predicting demand as a number preserves magnitude, while classifying demand as normal or high aligns directly with an escalation decision. The choice affects the loss, metric, calibration requirement, and action. Arbitrary discretization may discard useful information, whereas a continuous forecast may be harder to operationalize.

Ranking, survival, sequence, and structured prediction extend the framework. This lesson concentrates on regression and binary classification because they reveal the core logic. The correct task type follows from the target and decision, not from the analyst's preferred software.

2.6.4.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
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2.6.5 Professional Deepening 5 Data generating process and sampling

A model learns from the data-generating process represented by the sample. If the training sample differs systematically from the deployment population, ordinary test performance may be optimistic. Coverage error, nonresponse, convenience sampling, label selection, and institutional changes can all break representativeness.

Random sampling supports population inference when its assumptions hold, but operational data are often generated by service use rather than by a probability design. The absence of a record may mean no event, no access, no reporting, or a failed system. Analysts must map how each variable was created before interpreting it as a measurement.

Dependence also matters. Repeated observations from the same facility, household, district, patient, or device are not independent rows. A random row split can place nearly identical units in both training and test sets. Grouped, spatial, or temporal partitions are often necessary to reproduce the real deployment boundary.

2.6.5.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
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2.6.6 Professional Deepening 6 Features targets labels and prediction time

A feature is information supplied to the model. A target is the outcome against which predictions are evaluated. A label is the recorded operational representation of that target. These terms should not be used interchangeably: the intended construct and its recorded proxy may differ.

Prediction time is the most effective leakage-control device. For every candidate feature, ask whether its value would exist, be finalized, and be accessible when the model must issue the prediction. A field entered after case resolution is not a valid predictor of that resolution, even if it appears in the same database row.

Feature engineering should be specified as a reproducible transformation of permissible raw data. Missing-value treatment, scaling, category encoding, aggregation windows, and text processing must be learned from training data and then applied unchanged to validation, test, and production cases.

2.6.6.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
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2.6.7 Professional Deepening 7 Training validation and test partitions

Training data estimate model parameters. Validation data support choices among models, features, thresholds, and hyperparameters. Test data provide a final estimate after the selection process is complete. These roles are logical, not merely file names.

For independent and identically distributed observations, a random split may be adequate. For forecasting, training must precede validation and test in time. For grouped data, entire groups should remain together. For small samples, cross-validation can provide more stable selection evidence, but a final untouched test set remains valuable when feasible.

Repeated examination of test results leaks information through analyst behavior. Each adjustment made after seeing the test score adapts the system to that test sample. The remedy is procedural: predefine the test policy, record experiments, and reserve final evaluation for a controlled gate.

2.6.7.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
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- Which alternative design would give a more credible estimate of generalization?
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2.6.8 Professional Deepening 8 Baselines and incremental value

A baseline is the simplest credible alternative available at decision time. In regression it may be the training mean, a seasonal value, or the last observation. In classification it may be the majority class, a prevalence-based probability, or an existing rule.

The learned model must outperform the baseline on metrics aligned with the decision and under the same evaluation design. Improvement should be reported in absolute and relative terms. A small average improvement may still be valuable if it occurs in critical high-cost cases, but it may be immaterial if it disappears after uncertainty is considered.

Baselines also detect broken pipelines. If a complex model performs worse than a mean predictor, the problem may involve data preparation, target alignment, leakage removal, implementation error, or genuine absence of signal.

2.6.8.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
- Which information is unavailable, selectively observed, delayed or generated after the outcome?
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- What baseline, uncertainty measure and subgroup evidence must accompany the headline metric?
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2.6.9 Professional Deepening 9 Loss functions and optimization

Squared error,

$$L(y, \hat{y}) = (y - \hat{y})^2,$$

penalizes large regression errors strongly and leads to mean-oriented estimates. Absolute error,

$$L(y, \hat{y}) = |y - \hat{y}|,$$

is more resistant to extremes and targets a median under standard conditions. Classification log loss evaluates probabilities and heavily penalizes confident wrong predictions.

The training loss is an engineering objective, not automatically the final decision metric. A model may minimize log loss while the institution ultimately chooses a threshold based on capacity and error costs. Conversely, optimizing accuracy on an imbalanced problem may produce a model that ignores the rare class.

Optimization finds parameters that perform well under the specified objective. It does not validate the target, population, or causal interpretation. Convergence is therefore necessary but never sufficient evidence of readiness.

2.6.9.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
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- What baseline, uncertainty measure and subgroup evidence must accompany the headline metric?
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2.6.10 Professional Deepening 10 Model capacity underfitting and overfitting

Model capacity describes the variety of patterns a hypothesis class can represent. A low-capacity model may underfit: both training and validation errors remain high because relevant structure cannot be captured. A high-capacity model may overfit: training error becomes very low while validation error deteriorates.

Bias and variance provide a useful conceptual decomposition. High bias reflects systematic approximation error. High variance reflects sensitivity to the particular training sample. Noise is the irreducible component under the available information. The decomposition is a reasoning tool rather than a license to diagnose models from one score.

Regularization, simpler features, more representative data, cross-validation, and early stopping can control variance. Greater capacity, better features, and reduced measurement error can address underfitting. The appropriate response depends on evidence from learning curves and error analysis.

2.6.10.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
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2.7 The SRAI Supervised Learning Workflow

The canonical workflow for this chapter is:

Decision → Unit → Prediction Time → Target → Features → Split → Baseline → Model → Verification → Validation → Interpretation → Action.

Each arrow represents an intellectual responsibility.

2.7.1 Decision to unit

Define the action and identify what one prediction concerns.

2.7.2 Unit to prediction time

State when the prediction is issued and what information exists at that moment.

2.7.3 Prediction time to target

Define an outcome that occurs after the prediction and can be measured consistently.

2.7.4 Target to features

Select information that is relevant, measurable and legitimately available.

2.7.5 Features to split

Construct partitions that reproduce the deployment boundary across time, groups or locations.

2.7.6 Split to baseline and model

Fit transformations and models using training data. Use validation data for selection. Keep the test set untouched.

2.7.7 Model to verification

Check that code, transformations, metrics and data flow implement the intended procedure.

2.7.8 Verification to validation

Ask whether performance evidence supports the intended population, period and decision.

2.7.9 Validation to interpretation and action

Explain what the predictions mean, disclose uncertainty and limitations, and combine the evidence with operational judgment.

Artificial intelligence can assist at several stages.

It cannot erase the need for any stage.

2.8 Training Validation and Test Data

The three partitions serve different purposes.

Training data estimate model parameters and learned preprocessing transformations.

Validation data support choices among features, model classes, hyperparameters and decision thresholds.

Test data provide a final estimate after the selection process is complete.

These are logical roles, not merely filenames.

For independent observations, a random split may be defensible. For forecasting, earlier periods should train the model and later periods should evaluate it. For repeated entities, groups may need to remain intact. For spatial transfer, entire locations may need to be held out.

Repeated tuning against test results quietly converts the test set into validation data.

Preprocessing creates another boundary. Imputation, scaling, feature selection and encoding must be learned inside the training process. Fitting them before the split allows evaluation data to influence the model indirectly.

Verification asks: Did the pipeline enforce these roles correctly?

Validation asks: Do these partitions adequately represent the future use?

Strong SRAI practice requires both.

2.8.1 Professional Deepening 11 Learning curves and diagnostic reasoning

A learning curve plots performance against training-set size or model complexity. If training and validation errors converge at an unsatisfactory level, the system is often capacity-limited or feature-limited. If training performance is strong but validation performance lags, the model may have high variance, leakage in an earlier experiment, or a distribution mismatch.

Curves should be computed through a pipeline that repeats all learned preprocessing within each training fold. Otherwise the diagnostic itself can leak information. Variability across folds or repeated samples should be shown, because a smooth mean line can hide unstable performance.

Diagnostics guide the next experiment. They do not prove a single cause. Analysts should combine curves with subgroup results, residual plots, feature-range checks, data audits, and knowledge of the operating process.

2.8.1.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
- Which information is unavailable, selectively observed, delayed or generated after the outcome?
- Which alternative design would give a more credible estimate of generalization?
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2.8.2 Professional Deepening 12 Data leakage taxonomy and controls

Target leakage occurs when a feature contains the target or a post-outcome consequence. Train-test contamination occurs when preprocessing, duplicate entities, or feature selection uses information across partitions. Temporal leakage occurs when future observations influence past predictions. Group leakage occurs when related records appear on both sides of the evaluation boundary.

Leakage often looks like success: unusually high accuracy, implausibly clean residuals, or dramatic gains from a variable with a vague definition. Removing the leaking feature may cause performance to collapse, which is evidence that the earlier estimate was not operationally valid.

Controls include a prediction-time feature register, immutable split identifiers, pipeline-based preprocessing, duplicate and entity checks, temporal cut-offs, and independent review of high-importance features. Leakage review should occur before model comparison and again before deployment.

2.8.2.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
- Which information is unavailable, selectively observed, delayed or generated after the outcome?
- Which alternative design would give a more credible estimate of generalization?
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- What result would cause the analyst to stop, redesign the experiment or restrict intended use?

2.8.3 Professional Deepening 13 Regression metrics and residual analysis

Mean absolute error is expressed in target units and is easy to communicate. Root mean squared error,

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2},$$

gives extra weight to large errors. The coefficient of determination compares squared error with a constant baseline but can be negative on unseen data.

No single aggregate metric describes all behavior. Residuals should be examined against predictions, time, location, key features, and operational groups. Systematic residual patterns indicate missed structure, changing variance, omitted interactions, or data-quality problems.

Percentage errors are unstable near zero and can create misleading priorities. Metric selection must account for scale, asymmetry of consequences, and the decisions made from the forecasts.

2.8.3.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
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2.9 Supervised Learning Invariants

An invariant is a condition that should remain true throughout a controlled experiment.

The principal partition invariant is

$$I_{\text{train}} \cap I_{\text{validation}} = \emptyset, I_{\text{train}} \cap I_{\text{test}} = \emptyset, I_{\text{validation}} \cap I_{\text{test}} = \emptyset.$$

For a probabilistic classifier,

$$0 \leq \hat{p}_i \leq 1.$$

For a confusion matrix,

$$TP + TN + FP + FN = n_{\text{test}}.$$

For a preprocessing pipeline, all learned transformation parameters must be functions of training data only.

For a temporal task, every feature timestamp must satisfy

$$t_{\text{feature}} \leq t_{\text{prediction}} < t_{\text{target}}.$$

Invariants do not prove that the model represents the deployment environment well.

They reveal internal inconsistency and invalid information flow.

2.9.1 Professional Deepening 14 Classification metrics thresholds and calibration

A probabilistic classifier produces a score or estimated probability. A threshold converts that quantity into an action. Sensitivity measures the fraction of actual positives detected; specificity measures the fraction of actual negatives correctly excluded. Precision measures the fraction of predicted positives that are truly positive.

Accuracy can be misleading when classes are imbalanced. ROC AUC summarizes ranking across thresholds, while precision-recall analysis focuses attention on the positive class. Neither selects the operational threshold. The threshold must reflect prevalence, intervention capacity, and the costs of false positives and false negatives.

Calibration asks whether cases assigned probability 0.7 experience the event roughly 70 percent of the time in comparable groups. A well-ranked but poorly calibrated model may be unsuitable for resource planning without recalibration.

2.9.1.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
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2.9.2 Professional Deepening 15 Uncertainty stability and sensitivity

A test score is an estimate based on one finite sample. Confidence intervals, bootstrap distributions, repeated cross-validation, and subgroup sample sizes help show how much the estimate could vary. The objective is not decorative uncertainty reporting but honest decision support.

Sensitivity analysis asks whether conclusions change under defensible alternatives: different split dates, feature definitions, missing-data rules, thresholds, metrics, or random seeds. A model whose apparent advantage depends on one arbitrary choice requires caution.

Stability also concerns parameters and predictions. Large coefficient changes across samples, unstable feature importance, or volatile subgroup performance may undermine trust even when the average score is acceptable.

2.9.2.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
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- What result would cause the analyst to stop, redesign the experiment or restrict intended use?

2.10 Reproducibility

A result is reproducible when another analyst can use the recorded materials and procedure to obtain the declared result.

At minimum, a supervised learning experiment records:

- data source and version;
- unit of analysis and target definition;
- prediction time and feature boundary;
- split identifiers and dates;
- preprocessing sequence;
- model and hyperparameters;
- software environment;
- random seed;
- baselines and metrics;
- outputs and verification checks.

The companion notebook uses

```
SEED = 42
rng = np.random.default_rng(SEED)
```

The seed does not make the synthetic service records true.

It makes the experiment repeatable.

Reproducibility is necessary for scrutiny, but it is not the same as validity.

2.11 Canonical Worked Example Service Demand

The companion notebook constructs a synthetic facility-month dataset to make the complete supervised learning process visible. It does not claim to reproduce any real agency's confidential data.

2.11.1 Prediction design

- Unit: one facility-month.
- Issue time: end of month t .
- Target: visits in month $t + 1$.
- Action: allocate temporary staffing.
- Seed: 42.
- Observations after lag construction: 1,440.

The controlled temporal split is:

Partition	Rows	Period
Training	864	January 2020 to December 2022
Validation	288	January 2023 to December 2023
Test	288	January 2024 to December 2024

Future months do not influence earlier model development.

2.11.2 Regression baselines

The training-mean baseline produces

$$M A E = 88.880, R M S E = 103.958.$$

The seasonal-naive baseline produces

$$M A E = 31.426, R M S E = 40.496.$$

The validation process selects ridge regression with $\lambda = 1$. On the untouched test period, it produces

$$M A E = 24.728, R M S E = 30.876, R^2 = 0.909.$$

The notebook verifies that the learned model improves upon both declared baselines under the same evaluation design.

2.11.3 Leakage demonstration

The notebook deliberately introduces a post-outcome variable. The invalid leaked model reports

$$MAE = 1.650, RMSE = 2.048, R^2 \approx 1.000.$$

These extraordinary results are not celebrated.

They trigger diagnosis.

The feature is unavailable at prediction time and must be excluded. Removing it restores an honest estimate of future performance.

2.11.4 Classification framing

The notebook also predicts whether next-month demand exceeds a threshold fixed from training data. Validation selects a probability threshold of 0.625.

On the untouched test period:

- prevalence is 0.302;
- accuracy is 0.889;
- precision is 0.802;
- recall is 0.839;
- F1 is 0.820;
- ROC AUC is 0.955;
- log loss is 0.275.

These metrics answer different questions. The operational threshold still depends on staffing capacity and the costs of missed high-demand months and unnecessary interventions.

2.11.5 Uncertainty and subgroup record

A paired bootstrap estimates the test MAE improvement over the seasonal baseline. The controlled 95 percent interval is

$$[4.288, 9.053],$$

with all 1,000 bootstrap replicates showing positive improvement.

Regional MAE ranges from 22.53 to 27.16 visits. This variation is not automatically evidence of unfairness or failure. It is evidence requiring contextual investigation.

2.11.6 Reproducibility record

- Random seed: 42.
- Dataset: synthetic facility-month service demand.
- Split: frozen temporal blocks.
- Regression selection metric: validation RMSE.
- Selected model: ridge regression, $\lambda = 1$.
- Test regression RMSE: 30.876.
- Classification threshold: 0.625.

- Test ROC AUC: 0.955.
- Leakage control: post-outcome feature rejected.
- Execution status: 20 code cells completed with zero errors.

A fixed seed supports repeatability. It cannot compensate for a wrongly defined target, an invalid split, unrecorded preprocessing or a deployment population unlike the evaluation period.

2.11.1 Professional Deepening 19 Integrated case study service demand

An agency must forecast next-month visits for each facility so regional managers can allocate temporary staff. One row represents a facility-month. At the end of month t , the system predicts visits in month $t+1$. Permissible features include prior visits, calendar indicators, catchment population, planned closures known at prediction time, and lagged staffing.

The target is finalized next-month visits. A random row split is rejected because adjacent months from the same facility would cross partitions and because future periods would influence model selection. The team uses earlier periods for training, a later block for validation, and the most recent block for final testing, while retaining facility identifiers for grouped error analysis.

The training mean and a seasonal-naive forecast are baselines. Candidate models are compared on MAE, RMSE, high-demand error, and regional residuals. The selected model must improve operationally relevant metrics, remain stable across cut-off dates, and pass leakage review. The staffing decision remains with managers, who receive predictions, intervals, known limitations, and an override mechanism.

2.11.1.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
- Which information is unavailable, selectively observed, delayed or generated after the outcome?
- Which alternative design would give a more credible estimate of generalization?
- What baseline, uncertainty measure and subgroup evidence must accompany the headline metric?
- What result would cause the analyst to stop, redesign the experiment or restrict intended use?

2.11.2 Professional Deepening 20 Laboratory protocol

The companion notebook implements a synthetic but realistic learning laboratory. It creates demand data, freezes split identifiers, fits baselines and pipelines, evaluates regression and classification tasks, visualizes learning curves, demonstrates leakage, estimates bootstrap uncertainty, examines subgroup performance, and produces a compact evidence table.

Learners should run the notebook from the first cell without changing execution order. They should then alter one design choice at a time and record the effect. Recommended experiments include changing the temporal cut-off, increasing noise, removing a key feature, changing polynomial degree, moving the classification threshold, and introducing then removing a leaked feature.

The laboratory is complete only when the learner can explain why a score changed and whether that change represents genuine information, overfitting, leakage, sampling variation, or a different decision objective.

2.11.2.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
- Which information is unavailable, selectively observed, delayed or generated after the outcome?
- Which alternative design would give a more credible estimate of generalization?
- What baseline, uncertainty measure and subgroup evidence must accompany the headline metric?
- What result would cause the analyst to stop, redesign the experiment or restrict intended use?

2.12 Supervised Learning in Official Statistics

Official statistics increasingly combines designed surveys, censuses, administrative records, geospatial information and predictive methods.

The chain may include:

- defining the target population;
- constructing frames and observation units;
- measuring concepts;
- selecting or integrating records;
- creating labels;
- weighting or calibrating data;
- fitting predictive models;
- quantifying uncertainty;
- protecting confidentiality;
- communicating fitness for use.

Every step contains statistical and institutional choices.

A model trained on administrative users may not represent people without access to the service. A crop classifier evaluated on convenient plots may not transfer to remote areas. A missing record may mean zero activity, delayed reporting or system failure.

Supervised learning does not replace sampling and measurement knowledge.

It makes that knowledge more necessary.

2.13 Supervised Learning in Artificial Intelligence

Many AI systems are trained through supervised or partly supervised objectives. They produce classifications, scores, forecasts, rankings and generated outputs.

Behind these outputs are:

- labelled or preference data;
- feature representations;
- hypothesis classes;
- loss functions;
- optimization procedures;
- validation protocols;
- thresholds and decision rules.

Yet greater capacity does not guarantee trustworthy generalization.

An AI system may learn shortcuts created by the data collection process. It may optimize a proxy that is weakly connected to the real objective. It may produce a persuasive explanation that does not describe the mechanism behind its prediction.

The purpose of supervised learning in SRAI is not to make AI appear more impressive.

It is to make predictive evidence more understandable, testable and accountable.

2.13.1 Professional Deepening 16 Subgroup performance fairness and access

Aggregate performance can conceal systematic harm. Evaluation should reflect groups relevant to service access, geography, institution, time, and data quality. Differences require interpretation: they may reflect sample size, measurement quality, structural inequality, or a model that transfers poorly.

Fairness is not reduced to equalizing one metric. The appropriate criteria depend on the decision, legal context, intervention, and affected people. Analysts should document who benefits, who bears errors, whether protected or proxy attributes are used, and how appeals or human review operate.

Removing a sensitive attribute does not remove its information when correlated proxies remain. Conversely, excluding it from evaluation can prevent detection of unequal performance. Governance requires a deliberate distinction between attributes used for prediction and attributes used for auditing.

2.13.1.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
- Which information is unavailable, selectively observed, delayed or generated after the outcome?
- Which alternative design would give a more credible estimate of generalization?
- What baseline, uncertainty measure and subgroup evidence must accompany the headline metric?
- What result would cause the analyst to stop, redesign the experiment or restrict intended use?

2.13.2 Professional Deepening 17 Reproducibility and evidence management

A reproducible experiment records data version, target definition, split identifiers, feature transformations, software environment, random seed, model parameters, metrics, and outputs. A notebook is evidence only when it runs from a clean environment and produces results without hidden state.

Pipelines should keep transformations and models together. Relative paths, deterministic seeds, explicit dependencies, and small synthetic fallback data improve portability. Generated outputs should identify the code and configuration that produced them.

Reproducibility does not guarantee correctness, but it makes review possible. Controlled releases should distinguish source notebooks, executed evidence notebooks, exported reports, and production code.

2.13.2.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
- Which information is unavailable, selectively observed, delayed or generated after the outcome?
- Which alternative design would give a more credible estimate of generalization?
- What baseline, uncertainty measure and subgroup evidence must accompany the headline metric?
- What result would cause the analyst to stop, redesign the experiment or restrict intended use?

2.13.3 Professional Deepening 18 Deployment monitoring and governance

Deployment changes the risk landscape. Input distributions can drift, relationships can change, labels can arrive late, and users can adapt their behavior to the model. Monitoring should cover data quality, feature ranges, missingness, performance when labels mature, calibration, subgroup outcomes, latency, and override patterns.

Every system needs an owner, intended-use statement, excluded uses, review schedule, escalation path, and retirement criteria. A model card or equivalent record should summarize data, evaluation, limitations, and governance decisions.

Human oversight must be operational rather than ceremonial. Reviewers need sufficient information, authority, time, and a way to record disagreement. High-risk cases may require abstention or routing to a specialist instead of forced automation.

2.13.3.2 SRAI professional review

- What precisely is being predicted, for whom, and at what time?
- Which information is unavailable, selectively observed, delayed or generated after the outcome?
- Which alternative design would give a more credible estimate of generalization?
- What baseline, uncertainty measure and subgroup evidence must accompany the headline metric?
- What result would cause the analyst to stop, redesign the experiment or restrict intended use?

2.13.4 Governance Stress Test From Model Score to Authorized Use

A regional authority proposes to use the service-demand classifier to trigger automatic transfers of temporary staff. The model performs well on the controlled test period, but three questions remain unresolved. First, final demand labels arrive six weeks after each month closes. Second, staffing decisions themselves influence later demand because additional capacity attracts users who might otherwise travel elsewhere. Third, smaller facilities have fewer observations, wider uncertainty and greater sensitivity to a handful of unusual months.

The model score cannot resolve these institutional questions. The deployment record must state whether the system is advisory or automatic, who reviews high-impact recommendations, how delayed labels affect monitoring, how interventions alter the data-generating process, and which facilities require abstention or specialist review. It must also define what happens when predictions conflict with local information that is not represented in the dataset.

A controlled pilot should therefore separate predictive evaluation from operational evaluation. Predictive evaluation asks whether the system forecasts the declared target under a frozen design. Operational evaluation asks whether using the forecast improves allocation, reduces harmful queues, avoids destabilizing transfers and remains acceptable across regions. These questions require different evidence and may require phased implementation, randomized or quasi-experimental comparison, qualitative feedback and explicit stop criteria.

Before authorization, the owner should approve an intended-use statement, excluded uses, decision threshold, capacity constraint, override procedure, monitoring cadence, incident route and retirement rule. The model should be versioned together with its feature definitions, training window, validation record and deployment configuration. A later model with a higher score must not silently replace the approved version.

Monitoring must be designed before launch. Input monitoring should detect missing fields, invalid ranges, category changes and shifts in feature distributions. Performance monitoring should begin only when sufficiently mature labels are available and should preserve the same unit, target and metric definitions used at approval. Operational monitoring should examine who receives interventions, how often reviewers override recommendations, whether service delays improve, and whether scarce resources are displaced from populations not represented by the headline metric.

Thresholds require two levels. A warning threshold should trigger investigation while the system continues under heightened review. A suspension threshold should stop automated or high-impact use when data integrity, calibration, subgroup performance or institutional conditions depart materially from the approved envelope. Both thresholds should identify the responsible reviewer, evidence window, response time and documentation required to resume use.

Retraining is not an automatic remedy for drift. New data may encode temporary disruption, policy change, altered measurement or feedback created by the model itself. Before retraining, the team should determine whether the target and intended use remain stable, whether labels are comparable, whether the evaluation boundary must change and whether historical data should be down-weighted or excluded. The replacement model must pass the same controlled evidence gate as the original model.

Communication completes governance. Decision-makers should receive predictions with units, issue date, horizon, uncertainty, threshold meaning and known limitations. A dashboard that displays only a

risk score encourages false precision. A controlled interface should distinguish observed facts, model estimates, recommended actions, reviewer decisions and realized outcomes.

2.13.4.1 Required decision record

- Intended users and affected populations are identified.
- Predictions, recommendations and final decisions are stored separately.
- Human reviewers have authority, time and contextual information.
- Overrides are recorded without treating disagreement as analyst failure.
- Delayed outcomes are handled through an explicit monitoring lag.
- Drift thresholds distinguish warnings from mandatory suspension.
- Subgroup sample sizes and uncertainty accompany comparisons.
- Material incidents trigger review of data, code, model and decision process.
- Retirement is possible when the target, institution or intervention changes.

The governance conclusion is not that uncertainty prevents action. It is that action should be proportionate to evidence, reversible where possible, monitored after deployment and assigned to accountable people.

2.13.5 Comparative Audit Case Agricultural Early Warning

An agricultural agency intends to predict district-level crop stress six weeks before harvest. Candidate inputs include sample-survey observations, rainfall estimates, vegetation indices, crop calendars, planted-area estimates and historical yields. The decision is whether to intensify field verification, prepare targeted support and revise production forecasts.

The unit of prediction must be fixed before data integration. A district-season target differs from a plot-level target aggregated after prediction. The issue date must precede both the field outcome and any administrative response that could enter the records. Satellite composites must be restricted to acquisition dates available by the issue date, while survey variables must reflect interviews and processing completed by that date.

Sampling creates a second evidence boundary. Survey plots may follow a probability design, whereas satellite pixels provide broad coverage with different measurement error. Administrative reports may arrive more quickly from accessible districts. Combining these sources does not automatically eliminate their biases. The validation design should preserve survey weights or design information where relevant, hold out future seasons, examine agroecological zones and report uncertainty arising from both sampling and prediction.

A credible baseline may use the previous season, a five-year district mean or a simple rainfall anomaly rule. The learned model must demonstrate incremental value over these alternatives under the same temporal evaluation. Average RMSE should be accompanied by error in high-risk districts, calibration of alert probabilities, false-negative consequences and stability across alternative seasonal cut-offs.

Leakage review should examine final harvest assessments, post-harvest assistance records, revised area estimates and remotely sensed products processed with later observations. A variable can be historically stored under an early date even though its final value incorporates information unavailable at the real forecast date. The feature register must record availability and finalization, not only database timestamps.

Deployment should begin as decision support. Agronomists and district statisticians review alerts together with contextual evidence such as flooding, conflict, pest outbreaks and planting delays. Their decisions and reasons are stored separately from model predictions. This makes it possible to study complementary value instead of treating either local judgment or model output as unquestionable.

The comparison with service demand reveals what transfers and what changes. Prediction time, leakage control, baselines, uncertainty and reproducibility remain essential in both settings. The sampling design, spatial dependence, seasonal transfer, label delay and consequences of intervention require domain-specific treatment. SRAI therefore preserves a common evidence discipline without pretending that one technical recipe fits every operating system.

2.13.5.1 Audit questions

- Does every input exist and reach its controlled form by the forecast issue date?
- Does the evaluation reproduce transfer across seasons, districts and agroecological conditions?
- Are probability-sample and non-probability data sources distinguished?
- Do metrics reflect missed high-risk districts as well as average error?
- Are model alerts, expert reviews, interventions and realized outcomes stored separately?
- What evidence would suspend use or require a new validation design?

2.14 Thinking Box 2.1

A predictive model reports 99 percent training accuracy.

Before describing it as successful, what else must you know?

Consider:

- the performance of a credible baseline;
- validation and untouched test performance;
- class prevalence;
- temporal, geographic and group dependence;
- evidence of target or preprocessing leakage;
- calibration and threshold policy;
- uncertainty and subgroup performance;
- intended use and consequences of failure.

2.15 AI Reasoning Clinic

2.15.1 AI statement

“The model achieves almost perfect accuracy, so it is ready for deployment.”

2.15.2 Evaluation

The statement is false.

An impressive score does not demonstrate that:

- the target represents the intended outcome;
- features were available at prediction time;
- training and evaluation data are independent where required;
- preprocessing was fitted only on training data;
- the test set remained untouched;
- the evaluation population matches deployment;
- probabilities are calibrated;
- subgroup consequences are acceptable;
- monitoring and human oversight exist.

2.15.3 Improved statement

“The reported score is a promising result under the stated evaluation design. We must still audit the information boundary, compare baselines, verify the pipeline, quantify uncertainty, assess subgroup and deployment validity, and confirm that the output supports the intended decision.”

2.16 The SRAI Evidence Chain

A trustworthy supervised learning result should support the following chain:

Decision → Population → Prediction Time → Data → Representation → Split → Baseline → Model → Checks → Generalization Evidence → Interpretation → Action.

If one link is missing, confidence in the conclusion should decrease.

2.17 SRAI Principle No. 31

A predictive model earns trust through evidence of generalization, not through mastery of its training data.

Generalization becomes credible when the prediction boundary is explicit, partitions reproduce intended use, leakage is controlled, baselines are surpassed, uncertainty and subgroup behavior are examined, and the complete experiment is reproducible.

2.18 Key Insight

Supervised learning is not primarily the ability to fit a powerful algorithm.

It is the discipline of defining a future prediction, controlling information flow, separating model development from evaluation, and determining what conclusions unseen-case evidence can support.

2.19 Chapter Summary

This chapter established the foundation for Book 2.

We learned that:

- a supervised example pairs features with a target;
- prediction time determines which information is legitimate;
- empirical fit and future risk are different;
- hypothesis class, loss and regularization shape learning;
- training, validation and test data serve distinct purposes;
- baselines reveal incremental value;
- underfitting and overfitting require diagnostic evidence;
- leakage can make invalid systems appear excellent;
- regression and classification metrics answer different questions;
- uncertainty and subgroup assessment qualify average performance;
- reproducibility preserves the evidence chain;
- decisions require context, governance and human responsibility.

2.20 Exercises

2.20.1 Level 1 Concepts

1. Distinguish a feature, target and recorded label.
2. Explain why prediction time must be declared.
3. Differentiate training fit from generalization.
4. State the distinct roles of training, validation and test data.
5. Explain why a reproducible model may still be invalid.

2.20.2 Level 2 Interpretation

6. A crop-failure model uses the final insurance payment as a feature. Diagnose the problem.
7. A classifier reports 95 percent accuracy where only 5 percent of cases are positive. Explain why accuracy is insufficient.
8. A model performs well nationally but poorly in two remote regions. Identify evidence needed before drawing conclusions.

2.20.3 Level 3 Mathematical formulation

9. Write empirical risk for squared-error regression.
10. Define invariants for three mutually disjoint partitions.
11. Formulate a regularized objective and explain the role of λ .

2.20.4 Level 4 Programming

12. Run the companion notebook with seed 42 and record the environment.
13. Replace the temporal split with a random row split and explain the change in interpretation.
14. Introduce a deliberately leaked feature and design a check that rejects it.
15. Produce a reproducibility manifest containing data, split, pipeline, model, environment and outputs.

2.20.5 Level 5 Professional research

16. Select a published prediction system and reconstruct its evidence chain.
17. Compare two AI-generated model evaluations. Assess target definition, leakage, baselines, metrics and reproducibility.
18. Design a validation and monitoring protocol for a model supporting a public-sector allocation decision.

2.21 Companion Notebook Laboratory

Use M2_N01_supervised_learning_foundations.ipynb to:

1. inspect the computational environment;
2. reproduce the synthetic facility-month dataset;
3. verify the frozen temporal split;
4. compare mean and seasonal-naive baselines;
5. select ridge regularization using validation data;
6. evaluate the selected model once on the test period;
7. reproduce the leakage demonstration;
8. evaluate a classification threshold;
9. inspect bootstrap and subgroup evidence;
10. verify the controlled outputs.

The laboratory is complete only when the reasoning is documented. Merely running every cell is not sufficient.

2.22 Reproducibility Checklist

- Notebook identifier recorded

- Unit, target and prediction time recorded
- Data source and version recorded
- Split dates and identifiers frozen
- Software versions recorded
- Random seed controlled
- Preprocessing fitted inside training
- Baselines evaluated under the same design
- Cells executed in order
- Verification assertions passed
- Uncertainty and subgroup results recorded
- Limitations connected to the intended decision

2.23 Author's Perspective

Across official statistics, agricultural analysis, evaluation and information systems, weak predictive work often begins before an algorithm is selected. The unit is ambiguous. The outcome is measured after the decision it is supposed to inform. Operational records are treated as though they were a representative sample. A convenient score replaces a substantive question.

Artificial intelligence can now generate pipelines, select models and produce polished interpretations with extraordinary speed. That speed increases the importance of controlling the prediction boundary and the evaluation design.

The analyst's responsibility is not to obtain the highest number on a screen.

It is to establish what the number means, how it was obtained, whether it generalizes, who may be affected, and what decision the evidence can reasonably support.

2.24 Looking Ahead

Chapter 2 develops linear regression as a transparent baseline for continuous prediction.

It will examine the linear model, ordinary least squares, conditional coefficient interpretation, residual analysis, numerical stability, prediction uncertainty and the difference between predictive association and causal effect.

2.25 Verified References and Further Reading

- Bishop, C. M. (2006). *Pattern Recognition and Machine Learning*. Springer.
- Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The Elements of Statistical Learning*. Springer.
- James, G., Witten, D., Hastie, T., Tibshirani, R., and Taylor, J. (2023). *An Introduction to Statistical Learning with Applications in Python*. Springer.
- Mitchell, T. M. (1997). *Machine Learning*. McGraw-Hill.
- Shalev-Shwartz, S., and Ben-David, S. (2014). *Understanding Machine Learning*. Cambridge University Press.
- National Institute of Standards and Technology (2023). *Artificial Intelligence Risk Management Framework 1.0*. <https://doi.org/10.6028/NIST.AI.100-1>

2.26 Student Assessment Chapter 1

1. Differentiate empirical risk from expected future risk.
2. Name four items required to define a prediction-time information boundary.
3. Why is an untouched test set insufficient if preprocessing was fitted before splitting?
4. Design one verification invariant for a classification experiment.
5. Interpret a model whose leaked RMSE is 2.048 but valid test RMSE is 30.876.

2.27 Instructor Solutions Chapter 1

1. **Answer:** Empirical risk is average loss on observed development data; expected risk concerns loss on relevant future cases drawn from the operating population.
Marks: 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.
2. **Answer:** Prediction issue time, feature timestamp, feature finalization status, and target occurrence time; access permissions and operational availability are also valid.
Marks: 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.
3. **Answer:** Evaluation information has already influenced learned transformations, contaminating the pipeline even though model fitting later uses only training rows.
Marks: 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.
4. **Answer:** Confirm predicted probabilities remain between zero and one, or verify that $T P + T N + F P + F N = n_{test}$ on the fixed test set.
Marks: 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.
5. **Answer:** Treat the leaked result as invalid because a post-outcome variable entered the features. Use 30.876 as the controlled estimate under the declared temporal test, subject to uncertainty and deployment limitations.
Marks: 4 for correct result; 3 for valid reasoning; 2 for partial structure; 1 for relevant attempt.

Publication Control Record

AUDIT OUTCOME PASS WITH CONTROLLED REWRITE. The chapter is suitable for owner review before production of downstream assets or public release.

Control area	Finding	Resolution
Source authority	The legacy source grouped Chapters 1–5 and gave Chapter 1 only brief treatment.	Rebuilt Chapter 1 as an independent SRAI production unit while preserving the legacy package unchanged.
Chapter structure	The earlier review candidate used a handbook architecture unlike Book 1 Chapter 1.	Reconstructed the canonical 27-part SRAI chapter sequence and publication controls.
Notebook alignment	The original notebook was short and unexecuted; the first candidate expanded it.	Retained the expanded laboratory and aligned every controlled numeric result with its executed outputs.
Mathematical notation	Formulas must remain readable and editable.	Preserved equations as native Word mathematical objects and verified rendered output.
Assessment alignment	Legacy questions and solutions did not correspond.	Rewrote five assessment questions and matched each solution exactly.
Cross-channel controls	Public URLs are not yet final.	Leave links out until GitHub, website and YouTube destinations are public and verified.

Canonical cross-channel facts

- Title: Supervised Learning Foundations.
- Production unit: PU-B02-C01.
- Notebook: M2_N01_supervised_learning_foundations.ipynb.
- Seed: 42.
- Synthetic observations: 1,440 facility-months.
- Temporal test period: January to December 2024.
- Selected regression model: ridge regression, $\lambda = 1$.
- Regression test RMSE: 30.876.
- Classification threshold: 0.625.
- Classification test ROC AUC: 0.955.
- Leakage example RMSE: 2.048, rejected as invalid.
- Notebook execution status: passed with zero errors.